

TakhleeqX – AI-Powered Autonomous Marketing Agent

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Session 2022-2026

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June, 2022

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Chapter 1

Introduction [AS PER SCOPE DOCUMENT]

1.1 Introduction

1.1.1 Background

Digital marketing has become an essential component of business strategy in the modern economy, enabling brands to reach large audiences, engage customers, and measure campaign effectiveness in near real-time. Advances in natural language processing, generative models, and automation frameworks have created new opportunities to apply artificial intelligence (AI) for creative tasks and decision-making in marketing. Nevertheless, many small and medium-sized enterprises (SMEs) face barriers to adopting advanced digital marketing: cost, complexity, and the rapid pace at which trends evolve. Existing solutions frequently address only isolated parts of the marketing pipeline — for example, automated email workflows, copy generation, or analytics dashboards — and therefore require substantial human effort to coordinate end-to-end campaigns.

1.1.2 Problem Context and Motivation

Most traditional marketing agencies are expensive and often rely on static playbooks that fail to adapt quickly to changing market dynamics. Off-the-shelf automation platforms and copywriting assistants (e.g., email automation tools or content generators) provide useful functionality but still depend on human expertise to plan strategy, post across channels, and interpret results. This fragmented approach leads to inefficiencies: slow campaign iteration, inconsistent brand messaging, and suboptimal budget allocation. Furthermore, the majority of pre-trained AI models and automation tools are not optimized

for localized language variants, regional trends, or domain-specific behavior (e.g., local food preferences), which reduces their effectiveness for many small businesses. These gaps motivate the need for an integrated, adaptive solution that can analyze trends, create tailored creatives, deploy campaigns, and optimize performance in a closed feedback loop.

1.1.3 Project Overview

This project, **TakhleeqX**, proposes an AI-driven autonomous marketing system that combines multiple specialized agents into a single orchestrated pipeline. The system integrates a Trend Analysis Agent, a Content Creation Agent, an Ad Strategy Agent, an Automated Posting Agent, and a Performance Analytics Agent to provide end-to-end campaign automation. LangChain is used as the orchestration layer to connect large language models, external APIs, and auxiliary tools so agents can reason, call services, and maintain contextual memory across cycles. By leveraging pre-trained LLMs and vision models for content generation while logging campaign outcomes for future fine-tuning, TakhleeqX aims to be both practical for an MVP and extensible for long-term improvements.

1.1.4 Significance and References

The significance of TakhleeqX lies in its goal to make advanced digital marketing accessible and cost-effective for SMEs and local businesses that cannot afford full-service agencies. By automating strategy, creative production, deployment, and analytics within a closed-loop system, the project seeks to increase campaign responsiveness, improve ROI, and reduce manual workload. The design draws on prior work in AI-assisted marketing and agent orchestration; relevant literature and resources have been consulted to inform the architecture and methodology [1].

1.2 Existing Solutions

Over the past decade, numerous AI-driven and automation-based marketing tools have emerged, each addressing different parts of the digital marketing workflow. These systems offer capabilities such as automated content generation, social media scheduling, email marketing automation, and campaign analytics. While they provide valuable features, most existing solutions operate in *isolated silos* and do not provide a fully autonomous, end-to-end marketing pipeline. This creates significant gaps for small and medium-sized businesses that require both affordability and holistic automation.

Tools such as Jasper AI focus primarily on content generation and creative assistance, whereas platforms like Hootsuite emphasize post scheduling and social media management. Similarly, ActiveCampaign excels at automation for email marketing and CRM workflows but lacks capabilities for AI-driven creative generation or autonomous decision-making. Although these tools are widely used, they require continuous human involvement for trend analysis, strategy formulation, and cross-platform deployment.

The proposed system, TakhleeqX, addresses these limitations by integrating multiple AI agents using LangChain to automate the *entire* marketing lifecycle: trend analysis, creative generation, strategy planning, posting, engagement, and performance analytics. Such a closed-loop and autonomous structure is not currently offered by fragmented market tools, highlighting the need for a more comprehensive and intelligent marketing assistant.

Table 1.1: Comparison of Existing Solutions

System Name	System Overview	System Limitations
Jasper AI	AI-powered content generation tool designed to create marketing copy, captions, blogs, and creative text. Supports templates and brand voice modeling for consistent content creation.	Does not provide trend analysis, campaign strategy, posting automation, or multi-platform engagement. Requires human oversight for planning and execution.
ActiveCampaign	Customer experience automation (CXA) platform offering email marketing, CRM management, automated workflows, and segmentation-driven communication.	Focuses primarily on email and CRM. Lacks creative content generation, trend detection, cross-platform posting, and end-to-end autonomous marketing.

1.3 Problem Statement

In the modern digital landscape, marketing has become essential for the survival and growth of businesses, yet most small and medium-sized brands struggle to keep up with evolving trends, customer expectations, and competitive advertising strategies. Traditional marketing agencies are often expensive, slow, and heavily reliant on manual processes, making them inaccessible to many emerging businesses. Existing marketing tools offer isolated functionalities such as content creation, social media scheduling, or email automation, but none provide a fully integrated, autonomous system that manages the

complete marketing cycle. As a result, businesses frequently miss opportunities, misallocate budgets, or fail to produce engaging content aligned with real-time trends.

This project aims to solve these issues by developing an end-to-end autonomous marketing system powered by AI agents using LangChain. The system combines trend analysis, creative generation, ad strategy planning, automated posting, and performance analytics into a single unified platform. By reducing manual effort and decision fatigue, the proposed solution enables brands to maintain a consistent and data-driven digital presence without relying on costly agencies.

Furthermore, the project addresses the technical challenge of integrating multiple AI models and automation workflows into a closed feedback loop, allowing the system to learn from performance data and continuously improve campaign outcomes. Solving this problem will contribute to the field by demonstrating how multi-agent AI systems can revolutionize digital marketing, enhance efficiency, and democratize access to intelligent advertising tools for smaller businesses. Ultimately, the system aims to make marketing more affordable, adaptive, and effective for all business types.

1.4 Scope

The scope of this project focuses on designing and developing an autonomous AI-driven marketing system that streamlines the complete digital marketing workflow for businesses. The project will cover the creation of multiple integrated AI agents, including a Trend Analysis Agent, Content Creation Agent, Ad Strategy Agent, Automated Posting Agent, and Performance Analytics Agent. These agents will work collaboratively in a closed-loop system to automate tasks such as trend mining, ad content generation, campaign planning, scheduling, and performance evaluation. The project will also include the implementation of a web-based interface that allows users to interact with the system, upload business details, and receive AI-generated marketing assets.

Furthermore, the project scope includes developing backend workflows using LangChain to orchestrate agent communication and ensure seamless task delegation among the system modules. Integration with major platforms such as social media posting tools, APIs, and basic analytics systems will also be part of the development. The system will support text, image, and short-form video content generation using AI models. Additionally, basic authentication, onboarding flows, and admin monitoring features will be implemented to ensure secure and manageable system use.

However, the project will not cover advanced enterprise-level marketing tools such as large-scale CRM systems, high-budget ad bidding optimization, or real-time sentiment monitoring across all platforms. The development will also exclude long-term post-deployment maintenance and large-scale load testing, as these fall outside the academic

scope. Deep integrations with proprietary ad platforms like Meta Ads Manager or Google Ads will be limited to conceptual or prototype-level implementations. Financial, legal, and compliance-related aspects of digital advertising will also remain outside the project boundaries.

Overall, the project will provide a functional prototype demonstrating the feasibility and value of an AI-powered autonomous marketing system, while excluding complex commercial-scale enhancements that are beyond the timeframe and scope of FYP.

1.5 Modules

1.5.1 Business Onboarding Module

This module enables businesses to register on the platform and provide key details about their services, menu, and goals.

1. Collects business details such as name, menu, services, and offers.
2. Allows users to define marketing goals (brand awareness, sales increase, customer engagement).
3. Stores business preferences for personalization of campaigns.

1.5.2 Trend Analysis Agent Module

This module analyzes real-time market data, social media activity, and competitor performance to identify trending topics and customer preferences.

1. Gathers data from social media, reviews, and competitor ads.
2. Identifies trending topics, high-demand products, and customer preferences.
3. Generates market insights and passes them to the content creation module.

1.5.3 Content Creation Agent Module

This module generates ad creatives such as captions, posters, reels, and other promotional material using AI models.

1. Produces ad creatives including captions, posters, reels, and menu cards.

2. Customizes content based on trend insights and business inputs
3. Supports text and image generation using AI models.

1.5.4 Ad Strategy Agent Module

This module designs optimized ad campaigns by selecting audiences, platforms, budgets, and strategies for maximum impact.

1. Selects the best audience segments (e.g., students, families, office workers).
2. Chooses platforms (Facebook, Instagram, WhatsApp) for campaign deployment.
3. Sets campaign parameters such as budget, duration, and bidding strategy.

1.5.5 Automated Posting Agent Module

This module automates the scheduling and posting of campaigns across platforms, ensuring timely promotions and customer engagement.

1. Schedules and posts campaigns across selected platforms.
2. Automates daily offers and time-specific promotions.
3. Integrates chatbot functionality for responding to customer queries via DMs or WhatsApp.

1.5.6 Performance Analytics Agent Module

This module tracks, analyzes, and visualizes campaign performance to measure ROI and improve future strategies.

1. Tracks campaign KPIs such as impressions, clicks, reservations, and conversions.
2. Applies sentiment analysis to detect customer satisfaction or negative feedback.
3. Generates dashboards and reports to visualize ROI and campaign effectiveness.

1.5.7 Admin Control Monitoring Module

This module provides administrators with full control over campaigns and ensures proper system monitoring.

1. Provides oversight of all running campaigns.
2. Allows pausing, resuming, or stopping campaigns if needed.
3. Maintains logs and alerts for errors or anomalies.

1.5.8 Authentication Security Module

This module ensures secure user registration, authentication, and controlled access to the system.

1. Manages user registration and login.
2. Provides role-based access (e.g., business user vs. system admin).
3. Ensures data security for stored customer and campaign information.

1.5.9 Database Management Module

This module manages structured and unstructured data related to businesses, campaigns, and customer feedback.

1. Stores structured data such as campaign details, performance metrics, and user profiles.
2. Stores unstructured data such as reviews, social media insights, and customer feedback.
3. Ensures consistency and reliability of stored information.

1.5.10 Integration API Module

This module connects the web app with external APIs and ensures smooth communication with AI/ML models.

1. Handles connectivity with external services (Meta Ads API, Google Ads API, WhatsApp Business API).

2. Manages interaction between the web app and AI/ML models.
3. Ensures smooth data flow between system modules and third-party platforms.

1.6 Work Division

The following table outlines the distribution of responsibilities among the project team members for the development of the Short Reel Generation and Automated Post Generation modules.

Table 1.2: Work Division for Current Modules

Name	Registration No.	Assigned Responsibilities
Ayyan Ahmad	22i-0540	• Lead development of the Short Reel Generation pipeline. • Implement video generation logic using AI models. • Integrate caption-to-video mapping for reels.
Ali Hashim	22i-0554	• Design and develop the Automated Post Generation engine. • Implement templates for posters, captions, and social posts. • Ensure formatting consistency across platforms.
Hamza Jaffer	22i-0583	• Integrate both modules into the LangChain workflow. • Manage agent-to-agent communication (content → reel → posting). • Perform testing, refinement, and debugging of the integrated system.

Chapter 2

Project Requirements [AS PER FYP1 MID REPORT]

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-case/Event Response Table/Storyboarding

The system under development, **TakhleeqX**, is an AI-powered autonomous marketing platform that automates campaign management across multiple digital platforms. Given its user interactivity (business owners, marketers, and administrators), a **use-case approach** is the most suitable representation for this system.

Actor	Use Case	Description
Business Owner	Create Marketing Campaign	Allows business owners to input campaign goals (e.g., increase sales, promote a dish) and automatically generates ads using the Content Creation Agent.
Business Owner	View Campaign Performance	Enables users to view real-time analytics such as ROI, engagement rates, and audience growth provided by the Performance Analytics Agent.
AI System Administrator	Configure AI Agents	Allows administrators to manage model parameters, update LangChain pipelines, and ensure optimal system operation.
Marketing Analyst	Analyze Market Trends	Uses the Trend Analysis Agent to extract insights from social media, reviews, and search trends to guide campaign direction.
Customer (End User)	Interact with Chatbot	Enables end users to interact with AI chatbots for inquiries, orders, or reservations, and receive instant personalized responses.
Content Manager	Approve Generated Content	Provides an interface for reviewing and approving AI-generated ads, posts, and visuals before publishing.

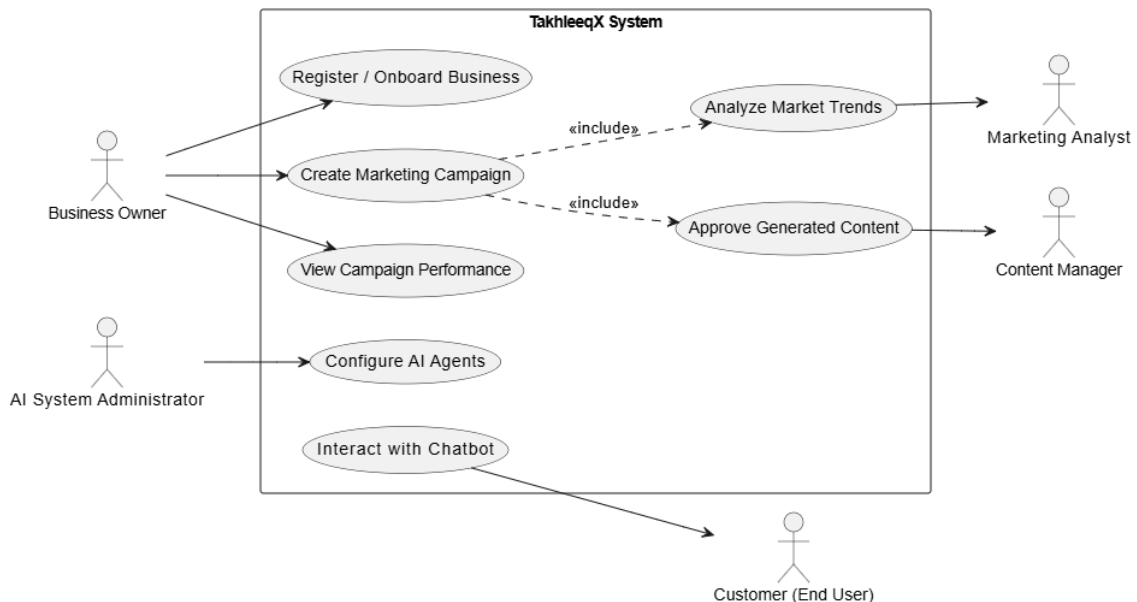


Figure 2.1: Use Case

Table 2.1: Primary Use Cases of TakhleeqX System

2.2 Functional Requirements

This section describes the key functional requirements of each module of the TakhleeqX system.

2.2.1 Module 1: Trend Analysis Agent

Following are the requirements for the Trend Analysis Agent:

1. FR1: The system shall collect trending topics and hashtags related to the restaurant industry from social media and review platforms.
2. FR2: The system shall analyze collected data to identify seasonal and location-based marketing opportunities.
3. FR3: The system shall summarize emerging customer interests using natural language generation for human review.
4. FR4: The system shall provide insights to the Ad Strategy Agent for real-time campaign adaptation.

2.2.2 Module 2: Content Creation Agent

Following are the requirements for the Content Creation Agent:

1. FR1: The system shall generate ad copies, captions, and hashtags based on the business type and target audience.
2. FR2: The system shall generate high-quality promotional images using text-to-image generation models.
3. FR3: The system shall maintain a consistent brand style across all visuals.
4. FR4: The system shall provide a preview option for owners to review content before publishing.

2.2.3 Module 3: Ad Strategy Agent

Following are the requirements for the Ad Strategy Agent:

1. FR1: The system shall select optimal advertising platforms (Facebook, Instagram, Google Ads) based on performance trends.

2. FR2: The system shall allocate budget dynamically using engagement prediction models.
3. FR3: The system shall automatically pause or boost ads based on real-time ROI analysis.
4. FR4: The system shall provide business owners with campaign comparison reports.

2.2.4 Module 4: Automated Posting Agent

Following are the requirements for the Automated Posting Agent:

1. FR1: The system shall schedule posts automatically across selected platforms.
2. FR2: The system shall handle post publication using integrated APIs (Facebook Graph, Instagram Basic Display).
3. FR3: The system shall respond to comments or direct messages using a conversational AI chatbot.
4. FR4: The system shall track user engagement per post for performance feedback.

2.2.5 Module 5: Performance Analytics Agent

Following are the requirements for the Performance Analytics Agent:

1. FR1: The system shall collect campaign performance data such as impressions, clicks, and conversions.
2. FR2: The system shall visualize KPIs through an interactive dashboard.
3. FR3: The system shall calculate ROI for each campaign and suggest data-driven adjustments.
4. FR4: The system shall export analytical reports in CSV and PDF formats for record keeping.

2.3 Non-Functional Requirements

This section specifies nonfunctional requirements. These quality requirements should be specific, quantitative, and verifiable. The following are some examples of documenting guidelines.

2.3.1 Reliability

The system shall maintain a minimum uptime of 99%. In the event of API failure or model malfunction, fallback routines shall switch to cached data and retry within 5 seconds. The system shall log all errors and automatically alert the administrator for manual inspection.

2.3.2 Usability

TakhleeqX shall feature a clean, dashboard-style interface that requires minimal training. All major actions (campaign creation, content review, analytics viewing) shall be accessible within three clicks. Onboarding tutorials and tooltips shall guide first-time users through the interface.

USE-1: The system shall allow a business owner to generate and launch a campaign within 5 minutes of registration.

2.3.3 Performance

PER-1: 90% of text and image generation operations shall complete within 10 seconds under normal network conditions (20 Mbps).

PER-2: The dashboard shall display updated campaign performance metrics within 30 seconds of data collection.

PER-3: Background tasks such as trend analysis and posting automation shall run asynchronously without blocking user interactions.

2.3.4 Security

SEC-1: All user credentials and business data shall be encrypted using AES-256 encryption.

SEC-2: API keys and tokens shall be stored in a secure vault and refreshed periodically.

SEC-3: The system shall include role-based access control to limit operations based on user roles (Admin, Analyst, Owner).

SEC-4: The system shall comply with standard data privacy practices to ensure no unauthorized data sharing.

Chapter 3

System Overview [AS PER FYP1 MID REPORT]

3.1 General Description

TakhleeqX is an AI-powered autonomous marketing system designed to revolutionize how small and medium-sized businesses manage their digital marketing. The system leverages multiple intelligent agents built using LangChain, Hugging Face local models, and generative AI frameworks to automate the entire marketing lifecycle—from trend research and content creation to advertisement deployment and performance analytics.

The primary objective of TakhleeqX is to eliminate the need for continuous human supervision in marketing operations. It achieves this by intelligently identifying trending topics and consumer interests through the **Trend Analysis Agent**, generating high-quality visuals and copy via the **Content Creation Agent**, developing optimized advertising campaigns through the **Ad Strategy Agent**, posting content automatically using the **Automated Posting Agent**, and monitoring campaign effectiveness through the **Performance Analytics Agent**. Each of these modules operates collaboratively to ensure an adaptive and self-optimizing marketing ecosystem.

The project operates in both online and offline modes, with Hugging Face-based local models enabling data privacy and offline functionality. This makes TakhleeqX ideal for businesses that prefer in-house automation without relying on cloud-based APIs. The system's modular architecture ensures scalability and extensibility, allowing for the integration of additional agents such as audience engagement analysis or budget optimization in future iterations.

By combining cutting-edge AI technologies with marketing intelligence, TakhleeqX aims to empower businesses to stay competitive in the digital landscape, reduce manual effort, and achieve data-driven decision-making with minimal human intervention.

3.2 Architectural Design

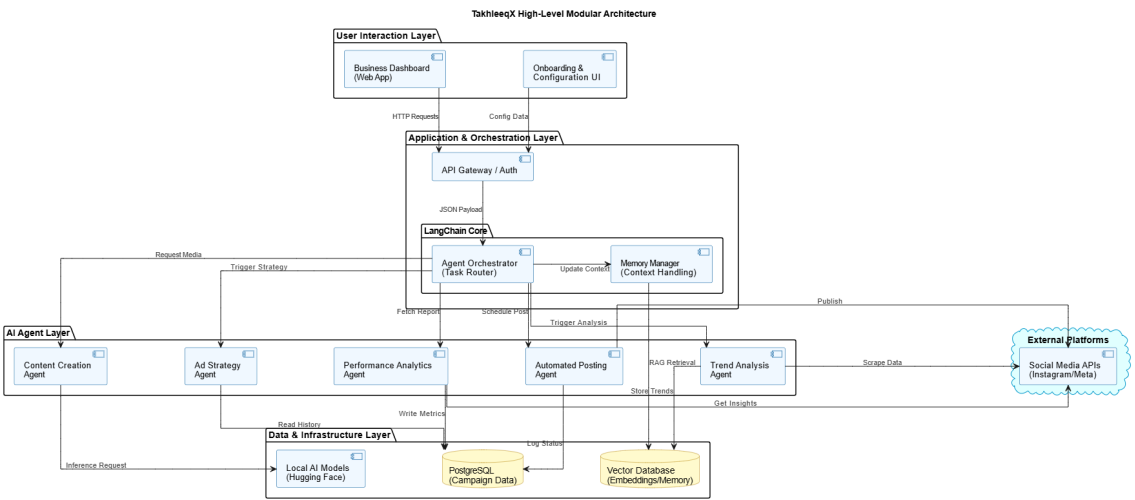


Figure 3.1: Architecture Diagram

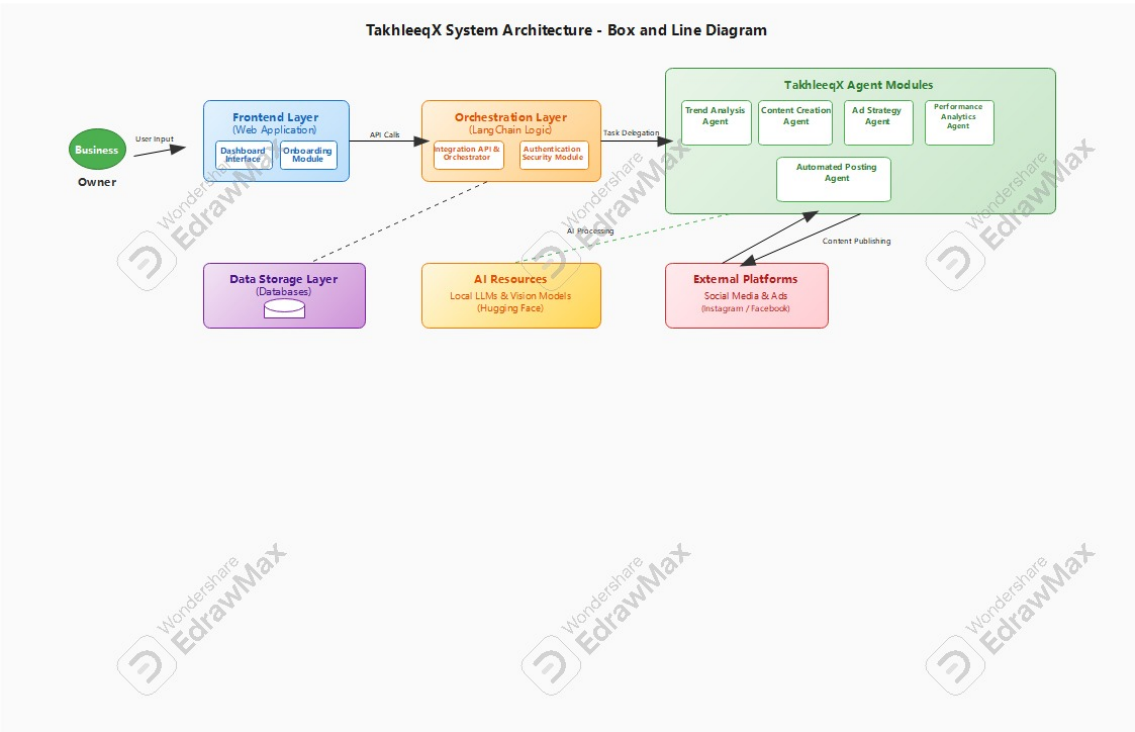


Figure 3.2: Box and Line

3.3 Data Design

Explain how the information domain of your system is transformed into data structures. Describe how the major data or system entities are stored, processed, and organized.

List any databases or data storage items.

3.4 Domain Model

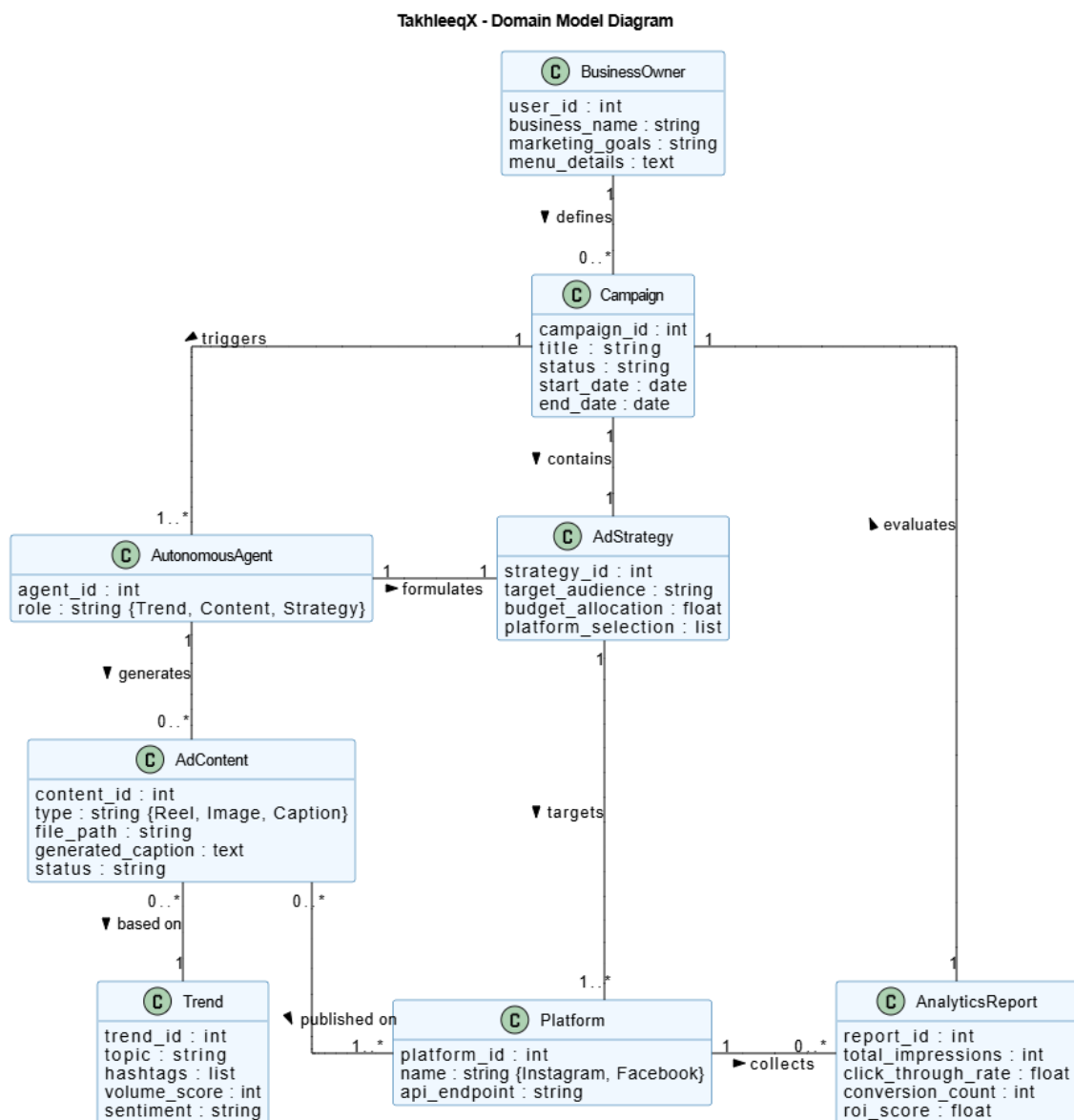


Figure 3.3: Domain Model

3.5 Design Models [UPTO THE CURRENT ITERATION ONLY]

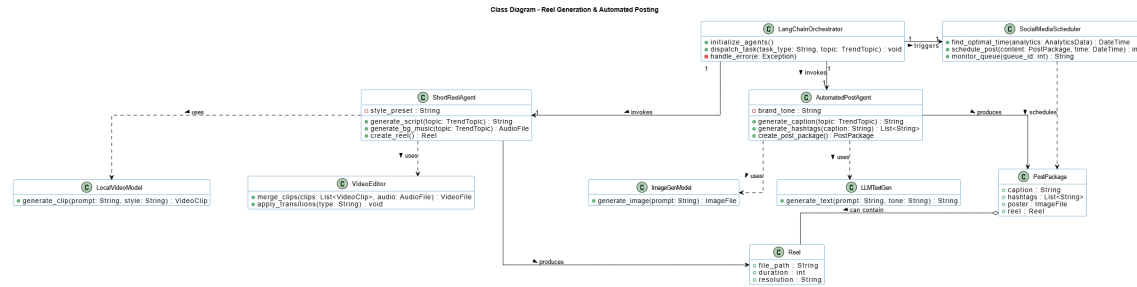


Figure 3.4: Class Diagram

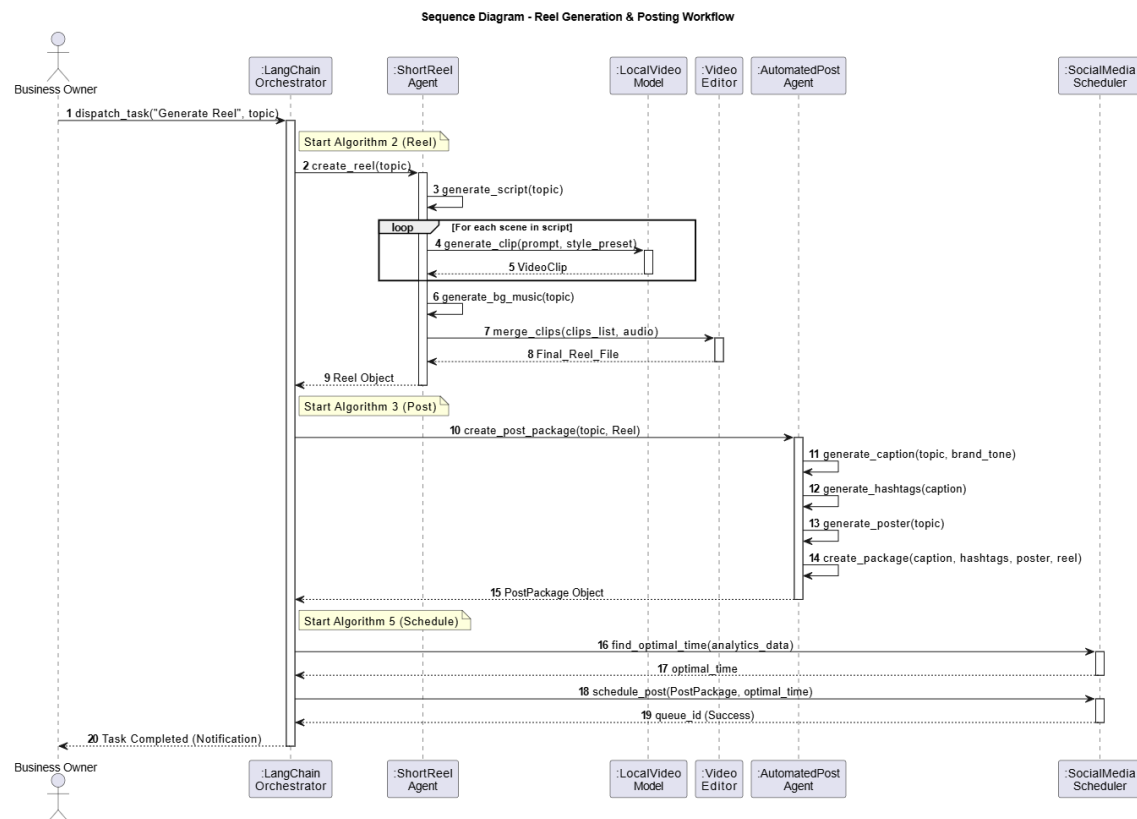


Figure 3.5: Sequence Diagram

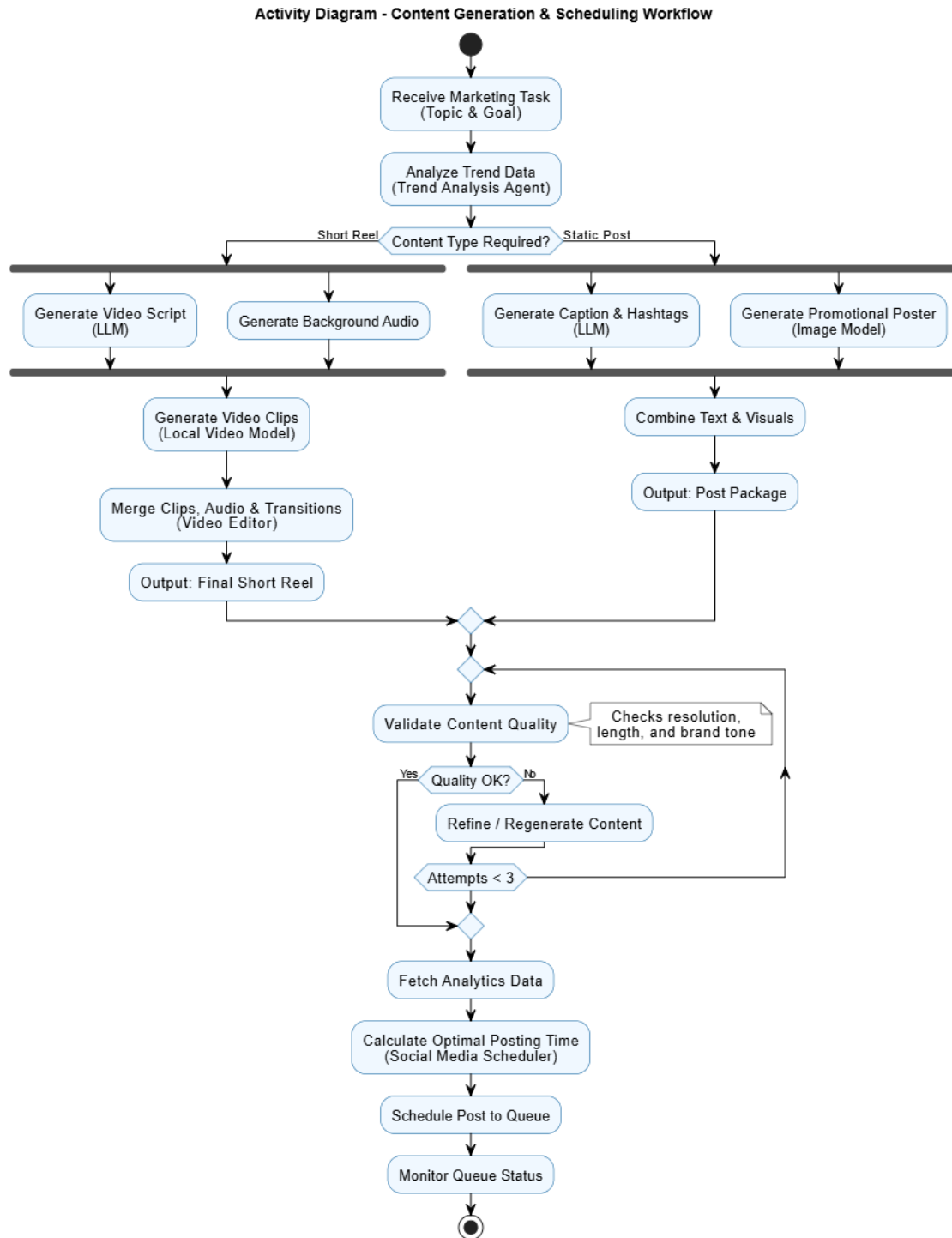


Figure 3.6: Activity

Chapter 4

Implementation and Testing [UPTO THE CURRENT ITERATION ONLY]

TakhleeqX is an AI-powered autonomous marketing system designed to transform how small and medium-sized businesses manage their digital marketing workflows. The system leverages multi-agent intelligence built using LangChain, locally hosted Hugging Face models, and generative AI frameworks to automate the entire marketing lifecycle. Its core purpose is to minimize human supervision while ensuring consistent, data-driven, and high-quality content production.

The architecture is composed of multiple intelligent agents, each responsible for a specific part of the marketing pipeline. The **Trend Analysis Agent** identifies emerging trends, consumer search patterns, and relevant content opportunities. The **Content Creation Agent** generates high-quality multimedia content—including short reels, ad creatives, and social media posts—using generative models and structured prompt templates. The **Ad Strategy Agent** formulates optimized advertisement plans, while the **Automated Posting Agent** handles cross-platform publishing without manual intervention. Additionally, the **Performance Analytics Agent** evaluates campaign performance and provides feedback that helps the system refine future content in a self-improving loop.

A key design aspect of TakhleeqX is its dual-mode operation. The system functions both online and offline, with local models ensuring privacy, cost reduction, and uninterrupted operation. This makes it ideal for businesses that prefer on-premise AI solutions without dependence on external APIs. The system follows a modular and scalable architecture in which each agent can be extended or replaced, enabling future enhancements such as budget optimization, engagement prediction, or multi-language content support.

By integrating advanced AI tools with automated marketing intelligence, TakhleeqX enables businesses to remain competitive in the digital landscape. It reduces manual workload, enhances campaign consistency, and supports data-backed decision-making. The

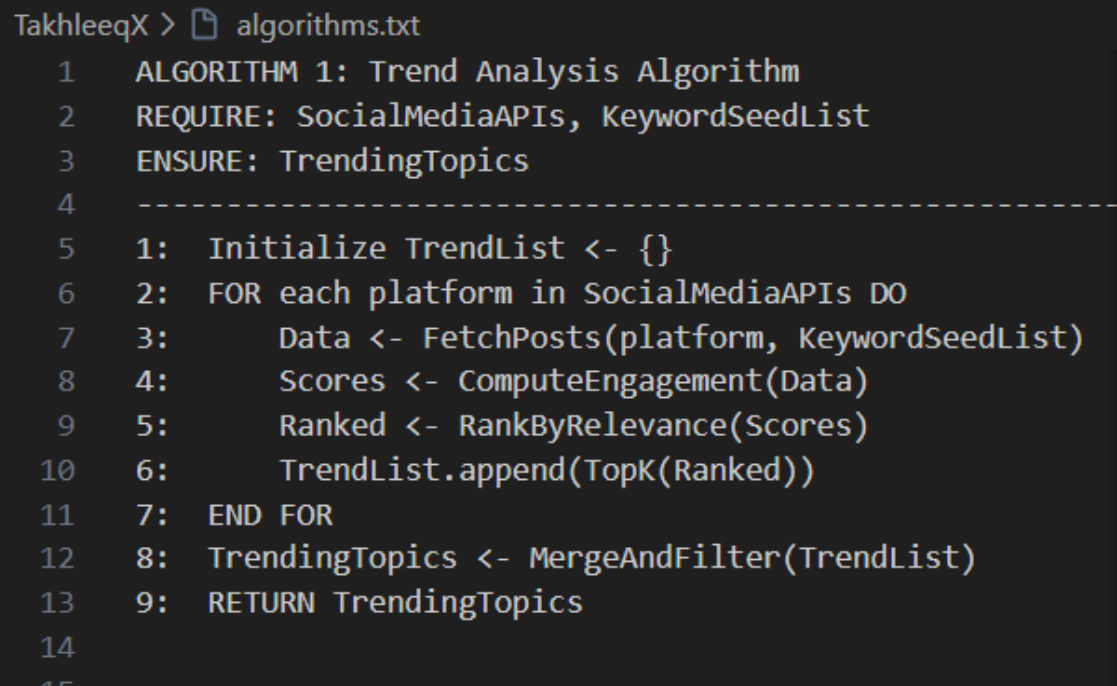
platform ultimately positions itself as a self-sustaining marketing assistant capable of operating with minimal human input while continuously adapting to trends and user needs.

4.1 Algorithm Design

This section outlines the core algorithms used in TakhleeqX across its major modules. Each module relies on AI-driven processes, including trend extraction, generative content creation, strategic ad generation, and automated scheduling. The pseudocode provided below explains the internal functioning of these components in a simplified and structured form.

4.1.1 Trend Analysis Algorithm

The Trend Analysis Agent identifies high-impact topics by scraping social media platforms, analyzing engagement scores, and ranking trends using weighted scoring.



```
TakhleeqX > algorithms.txt
1  ALGORITHM 1: Trend Analysis Algorithm
2  REQUIRE: SocialMediaAPIs, KeywordSeedList
3  ENSURE: TrendingTopics
4  -----
5  1: Initialize TrendList <- {}
6  2: FOR each platform in SocialMediaAPIs DO
7  3:   Data <- FetchPosts(platform, KeywordSeedList)
8  4:   Scores <- ComputeEngagement(Data)
9  5:   Ranked <- RankByRelevance(Scores)
10 6:   TrendList.append(TopK(Ranked))
11 7: END FOR
12 8: TrendingTopics <- MergeAndFilter(TrendList)
13 9: RETURN TrendingTopics
14
15
```

Figure 4.1: Algorithm 1

4.1.2 Short Reel Generation Algorithm

The Content Creation Agent uses generative AI to produce a short reel by combining text prompts, AI-generated clips, transitions, and background music.

```

ALGORITHM 2: Short Reel Generation Algorithm
REQUIRE: TrendTopic, StylePreset, LocalVideoModel
ENSURE: GeneratedReel
-----
1: Script <- GenerateScript(TrendTopic)
2: Clips <- LocalVideoModel.Generate(StylePreset, Script)
3: Audio <- GenerateBackgroundMusic(TrendTopic)
4: Reel <- Merge(Clips, Transitions, Audio)
5: RETURN Reel

```

Figure 4.2: Algorithm 2

4.1.3 Automated Post Generation Algorithm

This algorithm generates a complete social media post, including caption, hashtags, and optional poster image.

```

ALGORITHM 3: Automated Post Generation Algorithm
REQUIRE: TrendTopic, BrandTone, ImageModel, LanguageModel
ENSURE: PostPackage
-----
1: Caption <- LanguageModel.GenerateCaption(TrendTopic, BrandTone)
2: Hashtags <- GenerateHashtags(Caption)
3: Poster <- ImageModel.GenerateImage(TrendTopic)
4: PostPackage <- {Caption, Hashtags, Poster}
5: RETURN PostPackage

```

Figure 4.3: Algorithm 3

4.1.4 Ad Strategy Generation Algorithm

The Ad Strategy Agent analyzes audience segments, budgets, and platform suitability using a weighted decision mechanism.

```
ALGORITHM 4: Ad Strategy Generation Algorithm
REQUIRE: TrendInsights, HistoricalPerformance, Budget
ENSURE: OptimizedStrategy
-----
1: AudienceSegments <- IdentifyTargetGroups(TrendInsights)
2: PlatformScores <- EvaluatePlatforms(HistoricalPerformance)
3: AllocateBudget <- OptimizeBudget(AudienceSegments, Budget)
4: Strategy <- CreateAdPlan(AudienceSegments, PlatformScores, AllocateBudget)
5: RETURN Strategy
```

Figure 4.4: Algorithm 4

4.1.5 Automated Scheduler Algorithm

This agent schedules posts automatically by selecting the optimal posting time and queuing content for publishing.

```
ALGORITHM 5: Automated Posting and Scheduling Algorithm
REQUIRE: PostPackage, Platform, AnalyticsData
ENSURE: ScheduledPost
-----
1: BestTime <- FindOptimalTime(AnalyticsData)
2: QueueID <- Platform.Schedule(PostPackage, BestTime)
3: Monitor(QueueID)
4: RETURN ScheduledPost
```

Figure 4.5: Algorithm 5

4.2 External APIs/SDKs

Describe the third-party APIs/SDKs used in the project implementation in the following table. Few examples of APIs are provided in the table.

API/Library	Description	Purpose of usage	Main Class/Function used
LangChain	Framework for developing applications powered by language models	Orchestrating the flow between multiple agents (Trend, Content, Strategy)	LLMChain, PromptTemplate
Hugging Face Transformers	Open-source library for state-of-the-art NLP models	Loading and running local LLMs (DistilGPT2) for text generation	AutoModelForCausalLM, pipeline
Diffusers	Library for pretrained diffusion models	Generating marketing images from text prompts	StableDiffusionPipeline

4.3 Testing Details

Once the system has been successfully developed, testing has to be performed to ensure that the system working as intended.

4.3.1 Unit Testing

Each unit test is designed to test a specific function or method independently from other components, helping to identify issues directly related to the functionality being tested.

The following table outlines the unit tests performed on the core agents:

Test ID	Component	Test Description
UT-01	DataLoader	Verify that JSON data is loaded correctly and validates the 'post' field structure.
UT-02	TrendResearchAgent	Verify that the agent accepts a list of posts and returns structured JSON trend analysis.
UT-03	ContentAgent	Verify that the agent generates a caption string based on provided trend analysis inputs.
UT-04	ImageGenerator	Verify that the stable diffusion pipeline processes a prompt and saves a valid image file.

Table 4.1: Unit Test Cases for TakhleeqX Agents

4.3.2 Integration Testing

Integration testing ensures that all agents work together as a unified pipeline. The primary test case covers the end-to-end flow defined in the main execution script.

Test ID	Workflow	Expected Outcome
IT-01	End-to-End Pipeline	Data is loaded → Trends analyzed → Caption generated → Image created. All outputs must be present in the outputs/ directory.

Table 4.2: Integration Test Case

Bibliography

- [1] A Kolyshkin and S Nazarovs. Stability of slowly diverging flows in shallow water. *Mathematical Modeling and Analysis*, 2007.

Appendix A

Appendices

A.1 Appendix A

A.1.1 Use Case Diagram example (Online Shopping System)

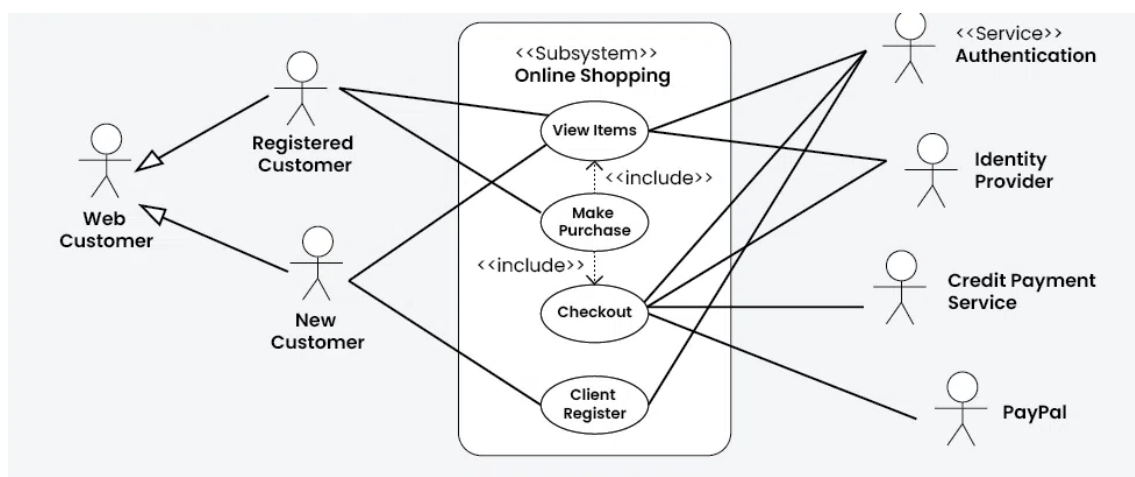


Figure A.1: Use Case Diagram for the Online Shopping System

A.1.2 Detail Use Case Example

ID	#1
Name	Overview elements
Short Description	All elements are shown in a list, where the user can set different filters.
Goal	Displaying elements in a changeable view.
Preconditions	The user got access to the system and is logged in. The user got the right to see schedules.
Success End Condition	The correct elements (filter and sorting) are displayed.
Fall End Condition	Elements not matching the criteria are displayed.
Stakeholder	Customer manager
Trigger	Login in as customer manager (because overview is on the starting screen for this role).
Normal Flow	1. The list of all elements matching default criteria are shown. 2. The user changes the filter criteria. 3. The user presses the Filter button. 4. The list shows all matching elements. Optional: 5. The user presses the Clear button. 6. The list of all elements matching default criteria are shown.
Alternative Flows	With click on the column headers, the list sorting can be changed. Different sorting for the different columns is described in the table below.
Includes	Login
Frequency of Use	About 50 times per day
Constraints and Special Requirements	None
Assumptions	None
Notes and Issues	None

Figure A.2: Detail Use Case Example

A.1.3 Event-Response Table for a Highway Intersection

Event	System State	Response
Road sensor detects vehicle entering left-turn lane.	Left-turn signal is red. Cross-traffic signal is green.	Start green-to-amber countdown timer for cross-traffic signal.
Green-to-amber countdown timer reaches zero.	Cross-traffic signal is green.	1. Turn cross-traffic signal amber. 2. Start amber-to-red countdown timer.
Amber-to-red countdown timer reaches zero.	Cross-traffic signal is amber.	1. Turn cross-traffic signal red. 2. Wait 1 second. 3. Turn left-turn signal green. 4. Start left-turn-signal countdown timer.
Pedestrian presses a specific walk-request button.	Pedestrian sign is solid Don't Walk. Walk-request countdown timer is not activated.	Start walk-request countdown timer.
Pedestrian presses walk-request button.	Pedestrian sign is solid Don't Walk. Walk-request countdown timer is activated.	Do nothing.
Walk-request countdown timer reaches zero plus the amber display time.	Pedestrian sign is solid Don't Walk.	Change all green traffic signals to amber.
Walk-request countdown timer reaches zero.	Pedestrian sign is solid Don't Walk.	1. Change all amber traffic signals to red. 2. Wait 1 second. 3. Set pedestrian sign to Walk. 4. Start don't-walk countdown timer.

Figure A.3: Example of Event Response Table

A.1.4 Story Board Example For Android App

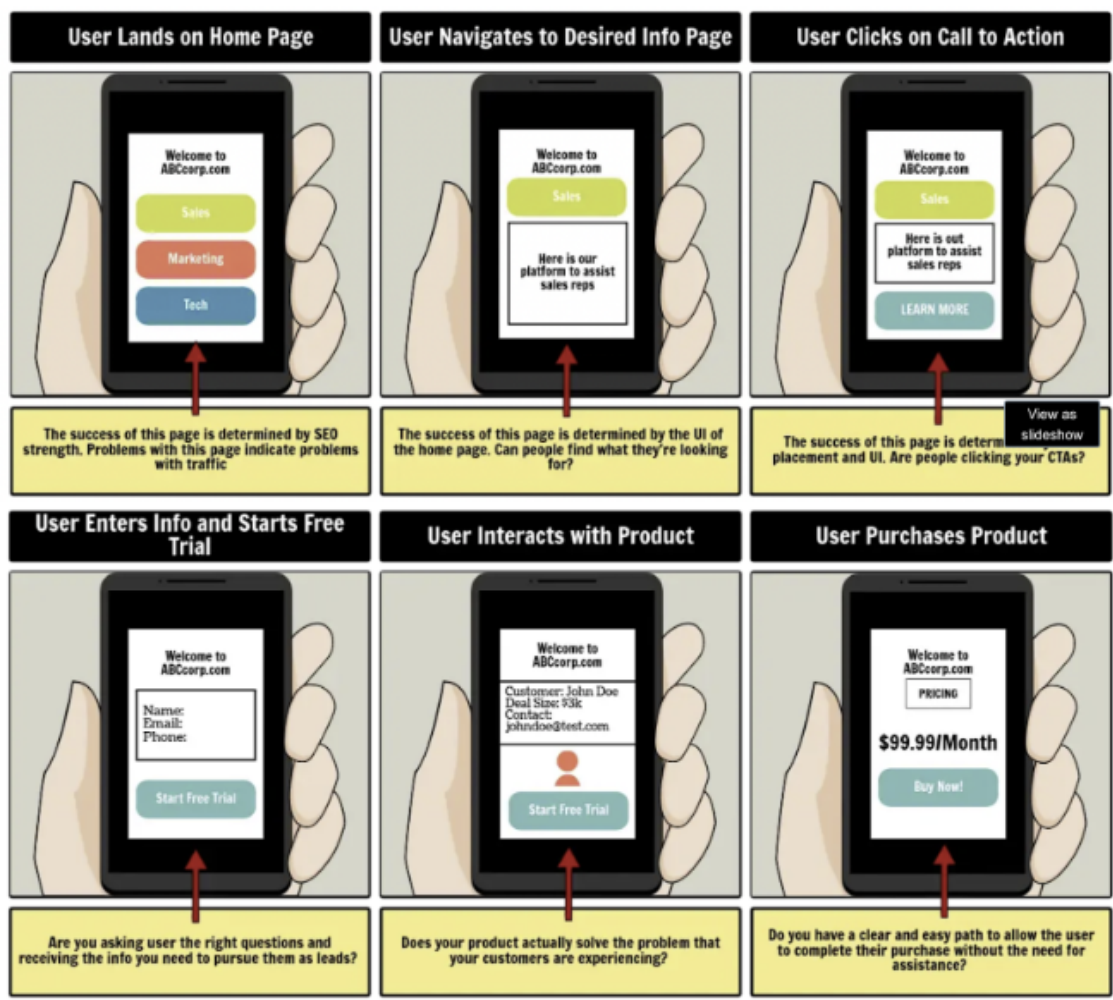


Figure A.4: Example of Story Board

A.2 Appendix B

A.2.1 Domain Model Example For Online Shopping

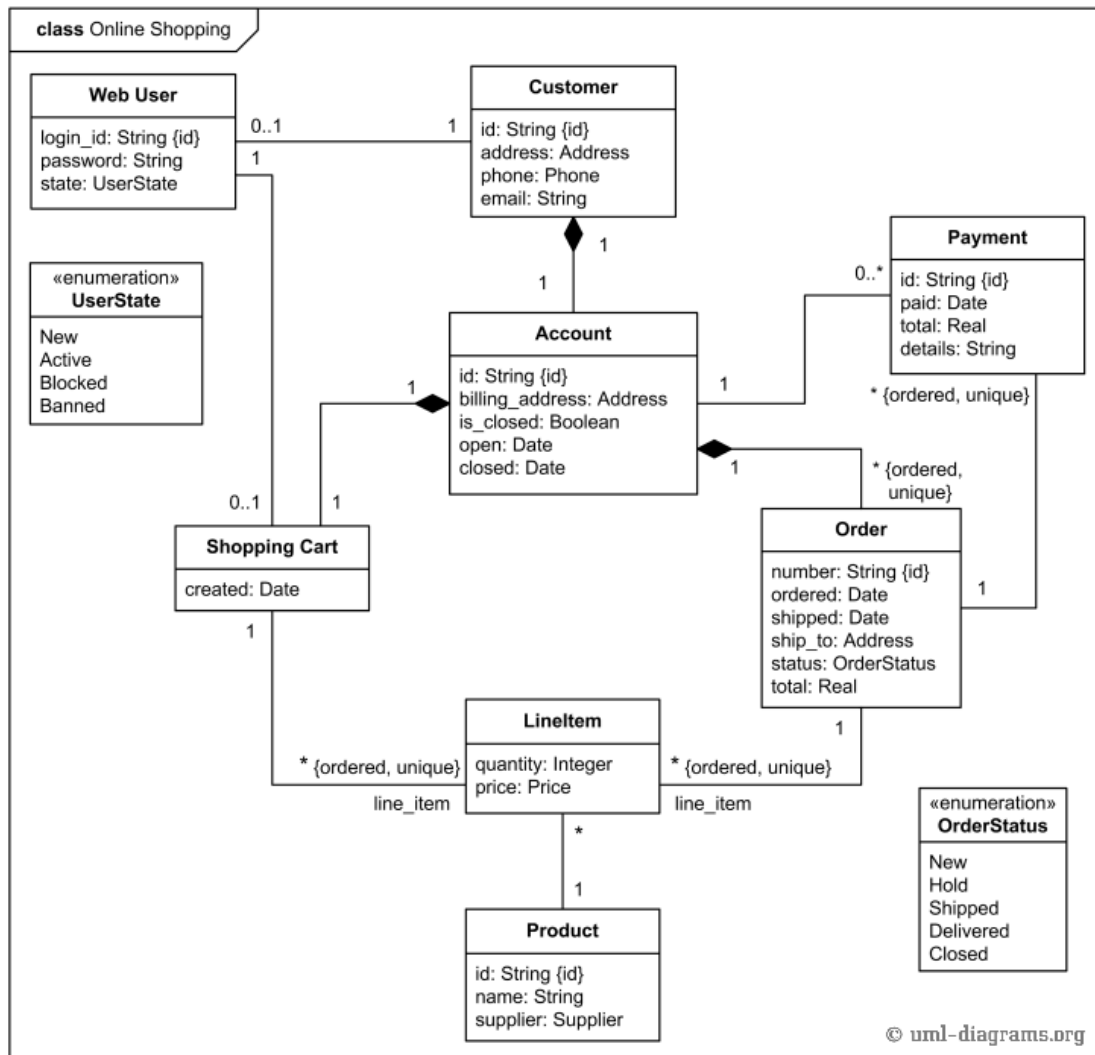


Figure A.5: Domain Model Example For Online Shopping Application

A.3 Appendix C

A.3.1 Box And Line Example For Online Shopping

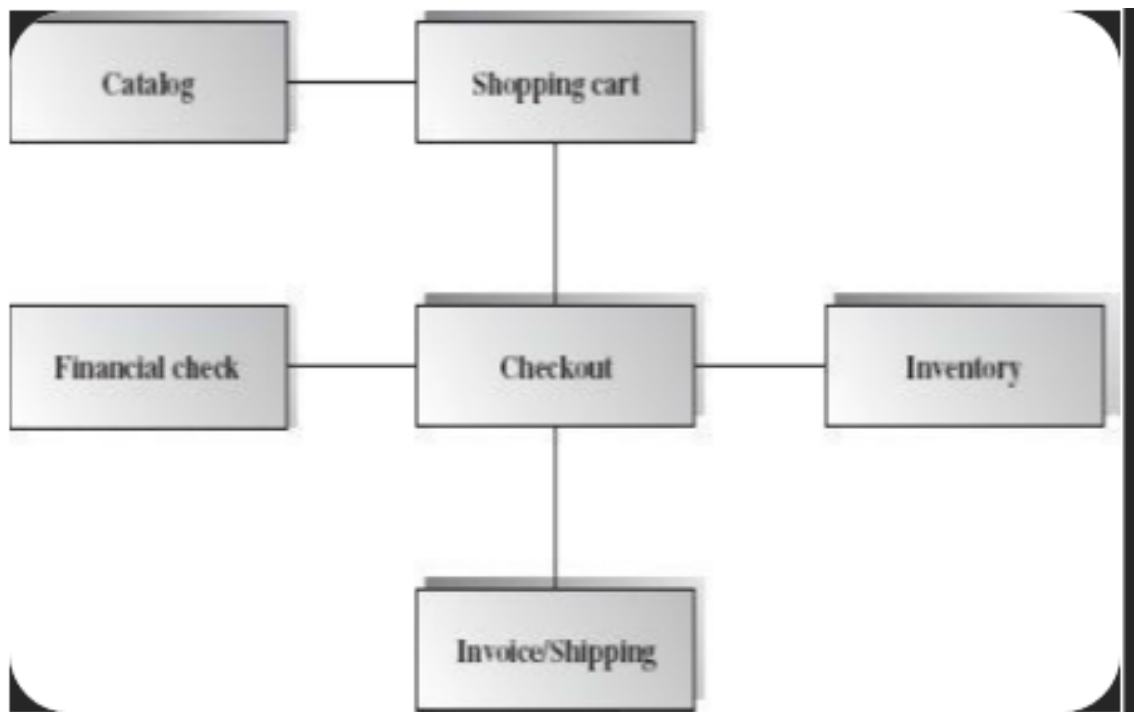


Figure A.6: Box and Line Diagram For Online Shopping Application

A.3.2 Architecture Pattern Example For Online Shopping

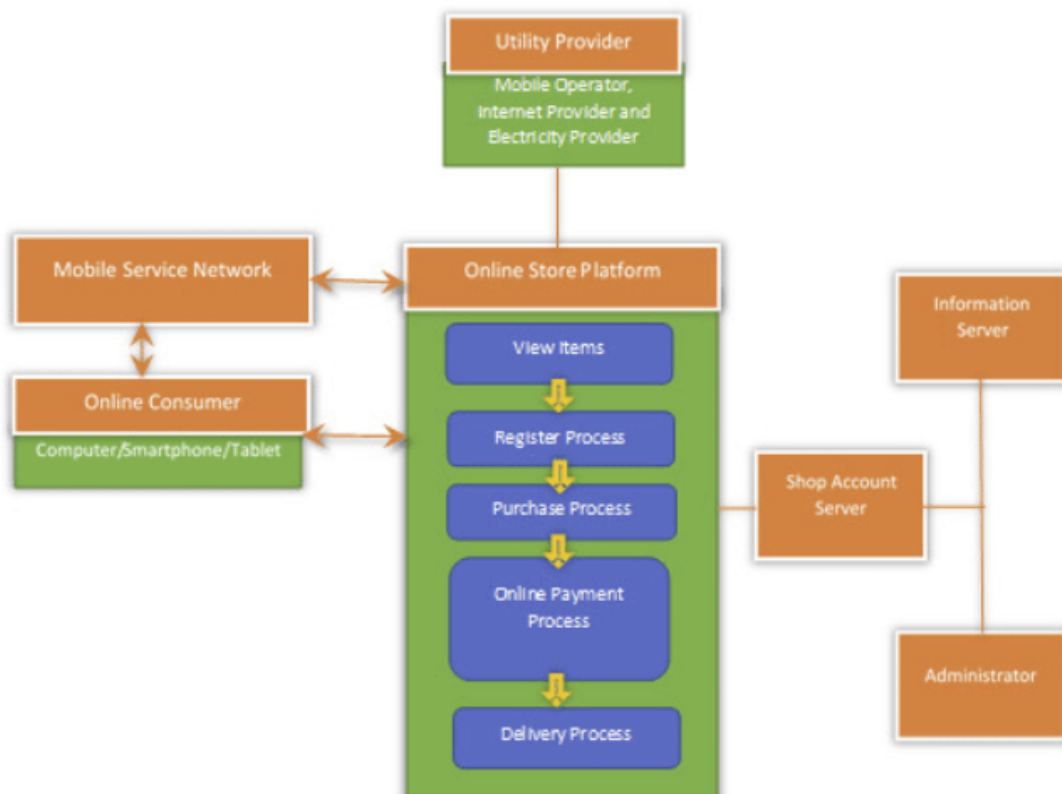


Figure A.7: Architecture Pattern For Online Shopping Application

A.4 Appendix D

A.4.1 Activity Diagram

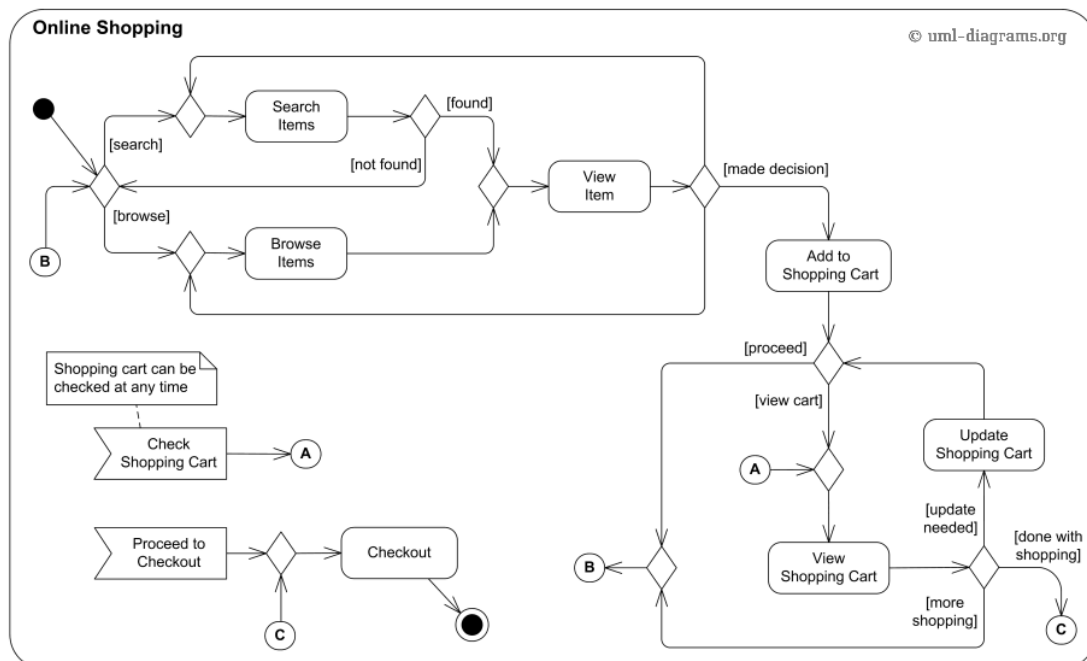


Figure A.8: Activity Diagram For Online Shopping Application

A.4.2 Class Diagram

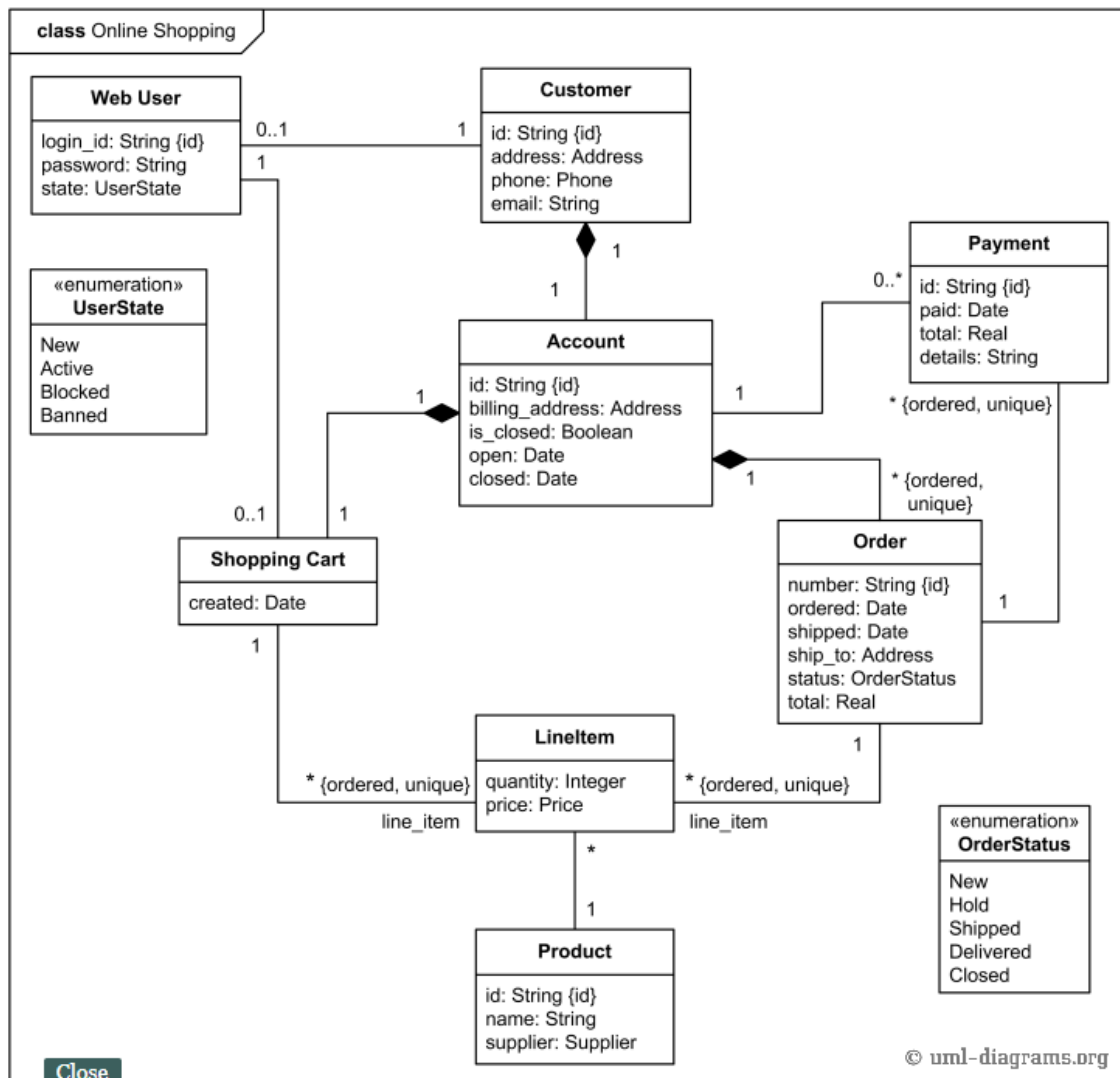


Figure A.9: Class Diagram For Online Shopping Application

A.4.3 Sequence Diagram

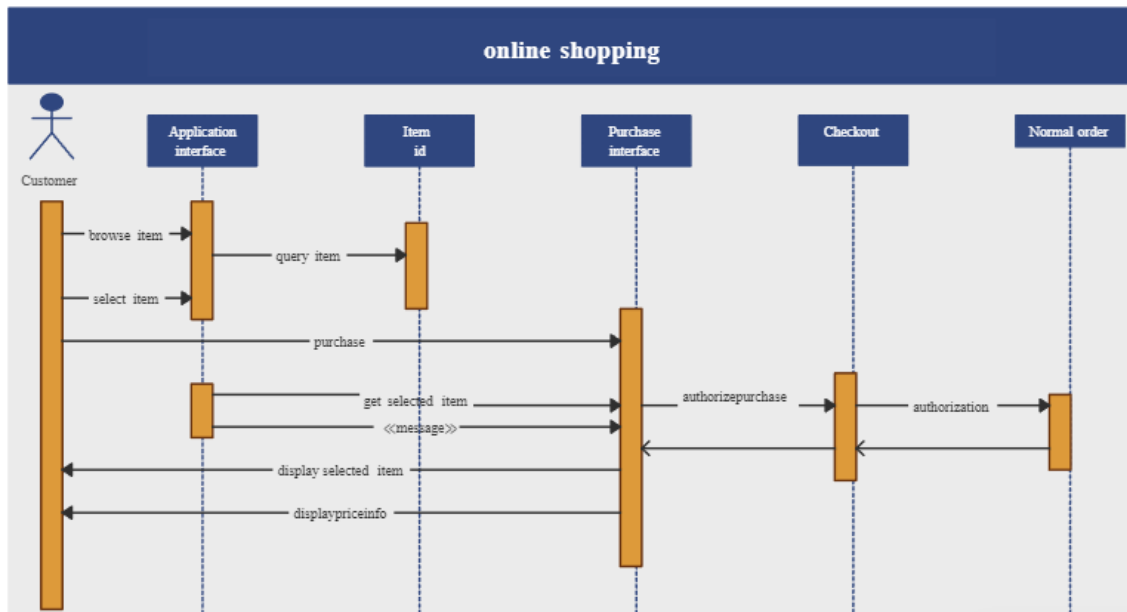


Figure A.10: Sequence Diagram For Online Shopping Application

A.4.4 State Transition Diagram

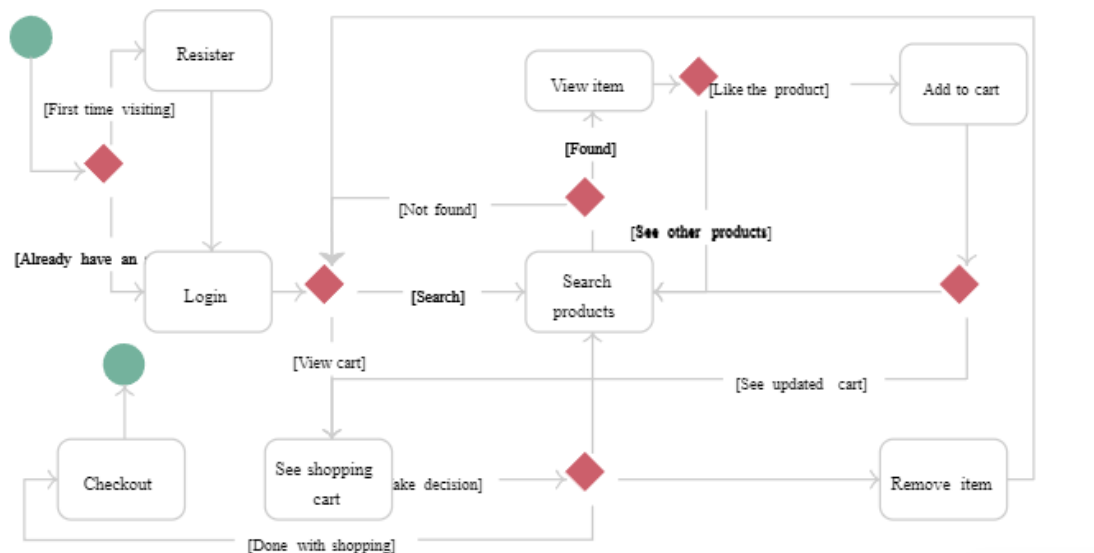


Figure A.11: State Transition Diagram For Online Shopping Application

A.4.5 Data Flow Diagram

Data Flow Diagram

Data flow diagram symbols, symbol names, and examples

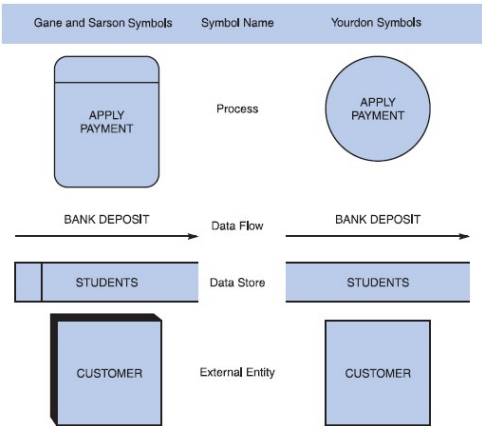


Figure B-5 Data flow diagram symbols, symbol names, and examples of the Gane and Sarson and Yourdon symbol sets.

Guidelines for Drawing DFDs

Step 1: Draw a Context Diagram: The first step in constructing a set of DFDs is to draw a context diagram. A **context diagram** is a top-level view of an information system that shows the system's boundaries and scope. Data stores are not shown in the context diagram because they are contained within the system and remain hidden until more detailed diagrams are created.

Example

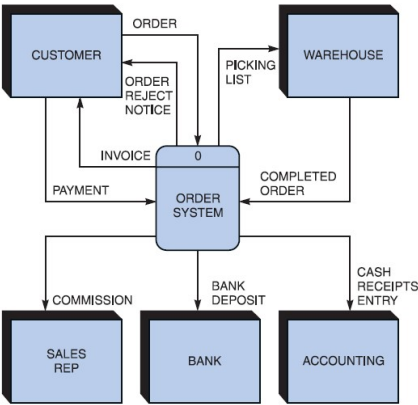


Figure B-6 Context diagram DFD for an order system.

Step 2: Draw a Diagram 0 DFD: To show the detail inside the black box, you create DFD diagram 0. **Diagram 0** zooms in on the system and shows major internal processes, data flows, and data stores. Diagram 0 also repeats the entities and data flows that appear in the context diagram. When you expand the context diagram into DFD diagram 0, you must retain all the connections that flow into and out of process 0.

Example

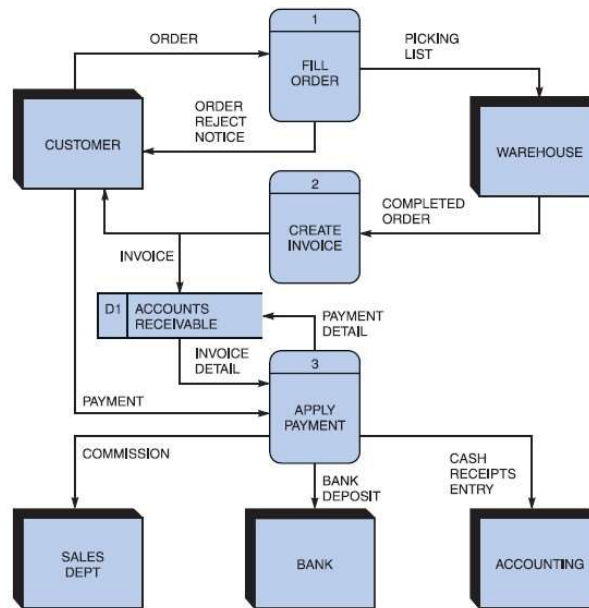


Figure B-7 Diagram 0 DFD for the order system.

Step 3: Draw the Lower-Level Diagrams:

To create lower-level diagrams, you must use leveling and balancing techniques. **Leveling** is the process of drawing a series of increasingly detailed diagrams, until all functional primitives are identified.

Leveling Example

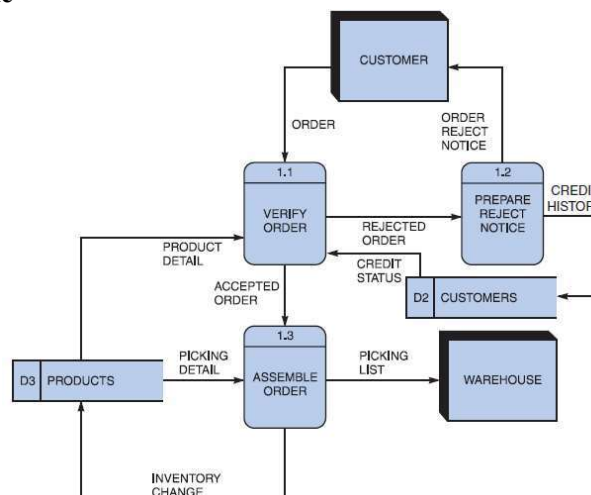


Figure B-8 Diagram 1 DFD shows details of the FILL ORDER process in the order system.